

# LUNA OS

THE AI-NATIVE CLINICAL COMPANION

*"Your body, finally understood."*

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# Introduction & Project Workflow

Luna OS is a high-performance AI companion engineered for women's health. Built specifically to handle complex biometric data and clinical symptoms, it ensures real-time, personalized care for the menopause transition.

## Biometric Ingestion

The platform captures live telemetry metrics like HRV (Heart Rate Variability), skin temperature, and sleep architecture via wearable integration.

## Background AI Operations

To guarantee an uninterrupted experience, heavy pattern-recognition (analyzing a 30-day longitudinal history of 34 symptoms) is offloaded to our background LLM engine.

# Problem Statement

## Complexity

A standard 12-minute doctor visit is completely insufficient to track and diagnose up to 34 distinct, interacting symptoms spanning over decades.

## Vulnerability

80% of OB-GYNs lack formal menopause training. Over 1.1 Billion women are affected globally, resulting in \$150B lost annually in workplace productivity due to unmanaged symptoms.

## Dependency

Existing FemTech platforms are merely menstrual trackers. They act as static lookup tables instead of natively analyzing complex perimenopausal transitions over time.

# Solution Approach

## Data Privacy (Local-First)

Isolates PII (Personally Identifiable Information) from the AI layer to maintain strict HIPAA-aligned safety protocols and prevent data collisions.

## Real-Time Safety Thread

Shifts emergency keyword classification to a prioritized loop. Instantly bypasses the AI and surfaces crisis lifelines if severe medical symptoms are detected.

## Context AI Engine

Injects the user's 30-day continuous symptom history directly into the LLM. Analyzes daily metrics against NAMS 2022 clinical guidelines to forecast trigger events natively.

# System Resilience

## Pattern Recognition (Concurrency)

Handles overlapping health inputs smoothly.  
Uses advanced RAG to cross-reference multiple clinical databases (vasomotor, sleep, psychiatry) instantly.

## Clinical Stability

Keeps AI outputs strictly evidence-based.  
Prevents AI hallucinations and medical over-reliance by refusing to recommend specific prescription dosages.

## Emergency Persistence

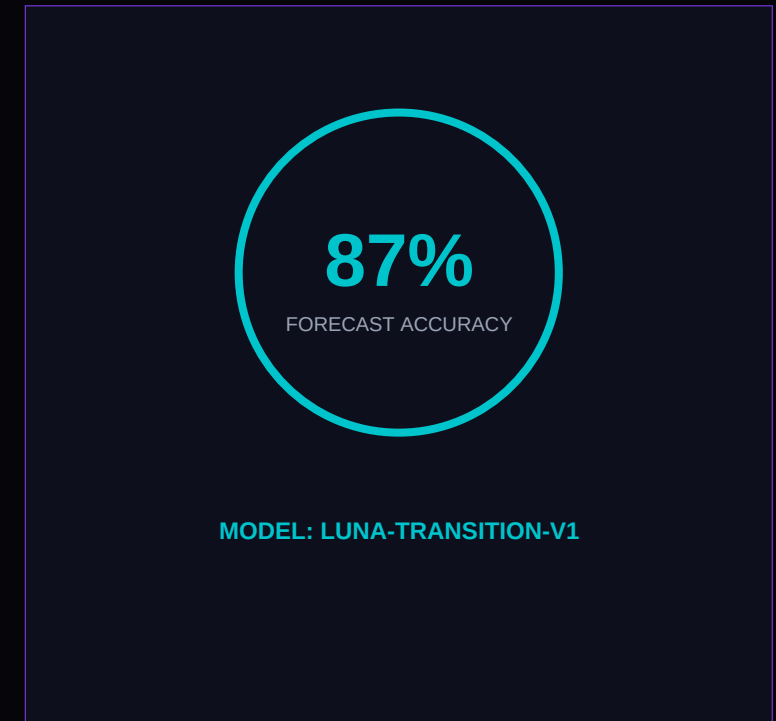
Enforces secure escalation paths. Any input matching crisis keywords triggers a hard-coded safety override.

SLIDE 06 // LIVE METRICS

# Clinical Intelligence

Luna OS integrates a native background predictive AI model to monitor, analyze, and forecast clinical transitions continuously.

- Luna OS integrates a native background predictive AI model.
- It monitors sleep architecture, daily mood logs, and biometrics continuously.
- The system automatically builds a real-time 24-Hour Hot Flash Risk Forecast.
- Generates a proactive "Doctor Prep" clinical dossier without manual input, saving vital time during physician visits.



# Architects, Core Stack & Roadmap

## THE ARCHITECTS

### Sonu Sharma

Lead Full-Stack & Systems Engineer

### Abdisa Jemal Hasen

Lead AI & Prompt Engineer

## CORE STACK

### Systems Frontend

React Native framework with high-contrast Tech-Noir custom Tailwind CSS design.

### Intelligence Plane

Anthropic Claude API for intent triage and RAG-grounded response generation.

### Backend Concurrency

Node.js API with PostgreSQL, running via Replit Agent for rapid live deployment.

## PROJECT ROADMAP

### P1 Hackathon Baseline

Deploying conversational AI & emergency triage safeguard.

### P2 Closed Beta

Onboarding 50 real users to validate trigger-detection.

### P3 Zenith Scalability

Stripe payment & B2B employer telehealth pilots.